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$$\begin{aligned} & \frac{1-2i}{(1+2i)^2} \\ &= \frac{1-2i}{-3+4i} \\ &= \frac{(1-2i)(-3-4i)}{(-3+4i)(-3-4i)} \\ &= \frac{(1-2i)(-3-4i)}{(-3)^2+(4)^2} \\ &= \frac{1 \cdot (-3) + 1 \cdot (-4i) - 2i \cdot (-3) + (-2i) \cdot (-4i)}{9+16} \\ &= \frac{-3-4i+6i+8i^2}{25} \\ &= \frac{-3+2i-8}{25} \\ &= \frac{-11+2i}{25} \\ &= -\frac{11}{25} + \frac{2i}{25} \end{aligned}$$

References